

DSA 8301: Statistical Inference for Big Data

Lecture Notes

Topic 3: Properties of Estimators

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3 Properties of Estimators

3.1 Unbiasedness

Definition 3.1 (Bias). For an estimator $\hat{\theta}$ of a scalar parameter θ , the bias is

$$\text{Bias}(\hat{\theta}) = \mathbb{E}_{\theta}[\hat{\theta}] - \theta.$$

$\hat{\theta}$ is unbiased if $\text{Bias}(\hat{\theta}) = 0$ for all θ .

3.1.1 Mean Squared Error (MSE).

Under squared-error loss $L(\hat{\theta}, \theta) = (\hat{\theta} - \theta)^2$, the corresponding *risk* is the mean squared error:

$$\text{MSE}_{\theta}(\hat{\theta}) = \mathbb{E}_{\theta}[(\hat{\theta} - \theta)^2].$$

Bias-variance decomposition. Write $\hat{\theta} - \theta = (\hat{\theta} - \mathbb{E}_{\theta}[\hat{\theta}]) + (\mathbb{E}_{\theta}[\hat{\theta}] - \theta)$. Then

$$\begin{aligned} \text{MSE}_{\theta}(\hat{\theta}) &= \mathbb{E}_{\theta}[(\hat{\theta} - \mathbb{E}_{\theta}[\hat{\theta}])^2] + (\mathbb{E}_{\theta}[\hat{\theta}] - \theta)^2 + 2 \underbrace{\mathbb{E}_{\theta}[(\hat{\theta} - \mathbb{E}_{\theta}[\hat{\theta}])]}_{=0} (\mathbb{E}_{\theta}[\hat{\theta}] - \theta) \\ &= \text{Var}_{\theta}(\hat{\theta}) + \text{Bias}(\hat{\theta})^2. \end{aligned}$$

Thus,

$$\text{MSE}_{\theta}(\hat{\theta}) = \text{Var}_{\theta}(\hat{\theta}) + \text{Bias}(\hat{\theta})^2.$$

If $\hat{\theta}$ is unbiased, MSE reduces to the variance.

Why MSE matters. Under squared-error loss $L(\hat{\theta}, \theta) = (\hat{\theta} - \theta)^2$, the *risk* is the mean squared error

$$\text{MSE}_{\theta}(\hat{\theta}) = \mathbb{E}_{\theta}[(\hat{\theta} - \theta)^2] = \text{Var}_{\theta}(\hat{\theta}) + \text{Bias}(\hat{\theta})^2.$$

MSE is central because it quantifies the fundamental tradeoff between variance (random fluctuation) and bias (systematic error).

- **Bias-variance tradeoff.** Any estimator must balance two sources of error:

$$\text{MSE} = \underbrace{\text{Var}(\hat{\theta})}_{\text{precision}} + \underbrace{(\mathbb{E}[\hat{\theta}] - \theta)^2}_{\text{systematic error}}.$$

Reducing variance often requires introducing a small bias (e.g., through shrinkage or regularization). Conversely, forcing unbiasedness can inflate variance, especially in small samples or ill-posed problems.

- **Among unbiased estimators, minimizing MSE $\iff$ minimizing variance.** If $\hat{\theta}$ is unbiased, then $\text{MSE}(\hat{\theta}) = \text{Var}(\hat{\theta})$. Hence the *best* unbiased estimator (in the MSE sense) is the one with the smallest variance. In regular parametric models this connects to the Cramér–Rao lower bound (CRLB):

$$\text{Var}_{\theta}(\hat{\theta}) \geq \frac{1}{n I(\theta)},$$

where $I(\theta)$ is Fisher information. Efficient unbiased estimators attain this bound.

- **Allowing small bias can substantially reduce variance and lower MSE.** A biased estimator can have *smaller* MSE than any unbiased competitor, especially at small n . Two illustrative calculations:

1. *Normal variance: biased MLE vs. unbiased estimator.* Let $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} N(\mu, \sigma^2)$ and

$$S_{\text{MLE}}^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2, \quad \hat{\sigma}^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2.$$

Then their MSE are computed as below

Let $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} N(\mu, \sigma^2)$ and define

$$\hat{\sigma}^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2, \quad S_{\text{MLE}}^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2 = \frac{n-1}{n} \hat{\sigma}^2.$$

Recall the chi-square fact:

$$\frac{(n-1)\hat{\sigma}^2}{\sigma^2} \sim \chi_{n-1}^2.$$

Chi-Square distribution

Claim. If $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} N(\mu, \sigma^2)$ and

$$\hat{\sigma}^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2,$$

then

$$\boxed{\frac{(n-1)\hat{\sigma}^2}{\sigma^2} \sim \chi_{n-1}^2}.$$

Step 1: Standardize. Define $Z_i = \frac{X_i - \mu}{\sigma} \sim N(0, 1)$ i.i.d., and let

$$\bar{Z} = \frac{1}{n} \sum_{i=1}^n Z_i, \quad \text{so that} \quad \bar{X} = \mu + \sigma \bar{Z}.$$

Then

$$\sum_{i=1}^n \frac{(X_i - \bar{X})^2}{\sigma^2} = \sum_{i=1}^n (Z_i - \bar{Z})^2.$$

Step 2: Sum-of-squares decomposition. Use the identity

$$\sum_{i=1}^n (Z_i - \bar{Z})^2 = \sum_{i=1}^n Z_i^2 - n \bar{Z}^2.$$

Since $Z_i \sim N(0, 1)$ i.i.d., we have

$$\sum_{i=1}^n Z_i^2 \sim \chi_n^2, \quad \sqrt{n} \bar{Z} \sim N(0, 1) \Rightarrow n \bar{Z}^2 \sim \chi_1^2.$$

Step 3: Orthogonalization (or Cochran's theorem). Let $Z = (Z_1, \dots, Z_n)^\top \sim N_n(0, I_n)$. Choose an orthogonal matrix A whose first row is $\frac{1}{\sqrt{n}}(1, \dots, 1)$. Then $U = AZ \sim N_n(0, I_n)$ with

$$U_1 = \sqrt{n} \bar{Z}, \quad U_2, \dots, U_n \text{ i.i.d. } N(0, 1), \text{ independent of } U_1.$$

Therefore

$$\sum_{i=1}^n (Z_i - \bar{Z})^2 = \sum_{j=2}^n U_j^2 \sim \chi_{n-1}^2,$$

and it is independent of $U_1 = \sqrt{n} \bar{Z}$ (equivalently, independent of $\bar{Z}$). This is a standard consequence of Cochran's theorem: the centered sum of squares is a chi-square with $n - 1$ d.f., independent of the sample mean.

Step 4: Conclude for the sample variance. Since

$$\frac{(n-1)\hat{\sigma}^2}{\sigma^2} = \frac{\sum_{i=1}^n (X_i - \bar{X})^2}{\sigma^2} = \sum_{i=1}^n (Z_i - \bar{Z})^2 \sim \chi_{n-1}^2,$$

the claim follows.

Corollaries.

- **Independence:** $\bar{X}$ and $\hat{\sigma}^2$ are independent in the normal model.
- **Moments:** Using χ_k^2 moments, $\mathbb{E}[\hat{\sigma}^2] = \sigma^2$, $\text{Var}(\hat{\sigma}^2) = \frac{2\sigma^4}{n-1}$.
- **Degrees of freedom:** One linear constraint (the sample mean) is estimated, leaving $n - 1$ orthogonal directions for variability around the mean, hence $n - 1$ d.f.

1) Mean and variance of $\hat{\sigma}^2$. Since $\mathbb{E}[\chi_k^2] = k$ and $\text{Var}(\chi_k^2) = 2k$,

$$\mathbb{E}[\hat{\sigma}^2] = \frac{\sigma^2}{n-1} \mathbb{E}[\chi_{n-1}^2] = \sigma^2, \quad \text{Var}(\hat{\sigma}^2) = \left(\frac{\sigma^2}{n-1} \right)^2 \text{Var}(\chi_{n-1}^2) = \frac{2\sigma^4}{n-1}.$$

Hence (unbiased):

$$\boxed{\text{MSE}(\hat{\sigma}^2) = \text{Var}(\hat{\sigma}^2) = \frac{2\sigma^4}{n-1}}.$$

2) Mean, variance, and bias of S_{MLE}^2 . Using $S_{\text{MLE}}^2 = \frac{n-1}{n} \hat{\sigma}^2$,

$$\mathbb{E}[S_{\text{MLE}}^2] = \frac{n-1}{n} \mathbb{E}[\hat{\sigma}^2] = \frac{n-1}{n} \sigma^2 \Rightarrow \text{Bias}(S_{\text{MLE}}^2) = \mathbb{E}[S_{\text{MLE}}^2] - \sigma^2 = -\frac{\sigma^2}{n}.$$

For the variance,

$$\text{Var}(S_{\text{MLE}}^2) = \left(\frac{n-1}{n}\right)^2 \text{Var}(\hat{\sigma}^2) = \left(\frac{n-1}{n}\right)^2 \frac{2\sigma^4}{n-1} = \frac{2\sigma^4(n-1)}{n^2}.$$

Therefore the MSE is

$$\text{MSE}(S_{\text{MLE}}^2) = \underbrace{\text{Var}(S_{\text{MLE}}^2)}_{\frac{2\sigma^4(n-1)}{n^2}} + \underbrace{\text{Bias}(S_{\text{MLE}}^2)^2}_{\frac{\sigma^4}{n^2}} = \sigma^4 \frac{2n-1}{n^2}.$$

$$\text{MSE}(S_{\text{MLE}}^2) = \sigma^4 \frac{2n-1}{n^2}.$$

3) Comparison ($\text{MSE}(S_{\text{MLE}}^2) < \text{MSE}(\hat{\sigma}^2)$ for all $n \geq 2$. Compare

$$\frac{2n-1}{n^2} \quad \text{vs} \quad \frac{2}{n-1}.$$

Since

$$\frac{2}{n-1} - \frac{2n-1}{n^2} = \frac{2n^2 - (2n-1)(n-1)}{n^2(n-1)} = \frac{3n-1}{n^2(n-1)} > 0 \quad (n \geq 2),$$

we have

$$\text{MSE}(S_{\text{MLE}}^2) < \text{MSE}(\hat{\sigma}^2) \quad \text{for all } n \geq 2.$$

Conclusion. Although S_{MLE}^2 is *biased* (downward), its mean squared error is *smaller* than that of the unbiased estimator $\hat{\sigma}^2$ for every $n \geq 2$. This is a canonical illustration of the bias–variance tradeoff: a small bias can reduce variance enough to lower overall MSE.

2. *Shrinkage of the mean (one-dimensional).* Suppose $X_i \stackrel{\text{i.i.d.}}{\sim} (\mu, \sigma^2)$ and estimate μ by $\tilde{\mu}_c = c \bar{X}$. Then

$$\text{Bias}(\tilde{\mu}_c) = (c-1)\mu, \quad \text{Var}(\tilde{\mu}_c) = c^2 \frac{\sigma^2}{n}, \quad \text{MSE}(\tilde{\mu}_c) = (c-1)^2 \mu^2 + c^2 \frac{\sigma^2}{n}.$$

Setup. Consider the shrinkage estimator $\tilde{\mu}_c = c \bar{X}$ for fixed μ . With $\text{Var}(\bar{X}) = \sigma^2/n$, write $\tau^2 := \sigma^2/n$. Then

$$\text{Bias}(\tilde{\mu}_c) = (c-1)\mu, \quad \text{Var}(\tilde{\mu}_c) = c^2 \tau^2,$$

so

$$\text{MSE}(c) = \underbrace{(c-1)^2 \mu^2}_{\text{squared bias}} + \underbrace{c^2 \tau^2}_{\text{variance}}.$$

Differentiation. Expand

$$\text{MSE}(c) = (\mu^2 + \tau^2)c^2 - 2\mu^2c + \mu^2.$$

Differentiate and set to zero:

$$\frac{d}{dc}\text{MSE}(c) = 2(\mu^2 + \tau^2)c - 2\mu^2 = 0 \implies c^* = \frac{\mu^2}{\mu^2 + \tau^2} = \frac{\mu^2}{\mu^2 + \sigma^2/n}.$$

Second derivative $2(\mu^2 + \tau^2) > 0$, so this is the unique minimizer.

Therefore

$$\text{MSE}(\tilde{\mu}_{c^*}) = \frac{\sigma^2}{n} \cdot \frac{\mu^2}{\mu^2 + \sigma^2/n} < \frac{\sigma^2}{n} = \text{MSE}(\bar{X}).$$

since

$$0 < \frac{\mu^2}{\mu^2 + \sigma^2/n} < 1$$

Thus, some (data-dependent) shrinkage can strictly improve MSE relative to the unbiased $\bar{X}$, especially when the signal-to-noise ratio $\mu^2/(\sigma^2/n)$ is small.

Decision-theoretic and practical implications.

- **Estimator choice under squared loss:** Pick the estimator with the smallest MSE at the θ values of interest; if θ is uncertain, consider average (Bayes) risk or uniform performance.
- **Regularization:** Ridge, lasso, and other penalized methods introduce bias to reduce variance, often yielding substantially lower prediction or estimation MSE.
- **Large samples:** As $n \rightarrow \infty$, many reasonable estimators become approximately unbiased and their MSEs are dominated by variance; asymptotic efficiency then governs comparisons.

3.1.2 Asymptotic Mean Squared Error (AMSE)

Definition 3.2. For an estimator $\hat{\theta}_n$ of a parameter θ , the mean squared error (MSE) is

$$\text{MSE}(\hat{\theta}_n) = \mathbb{E}[(\hat{\theta}_n - \theta)^2] = \text{Var}(\hat{\theta}_n) + (\text{Bias}(\hat{\theta}_n))^2.$$

The asymptotic mean squared error (AMSE) is defined by scaling the MSE with the sample size n and taking the limit:

$$\text{AMSE}(\hat{\theta}_n) = \lim_{n \rightarrow \infty} n \cdot \text{MSE}(\hat{\theta}_n),$$

provided the limit exists.

Expanding the definition gives

$$\text{AMSE}(\hat{\theta}_n) = \lim_{n \rightarrow \infty} \left[n \text{Var}(\hat{\theta}_n) + n (\text{Bias}(\hat{\theta}_n))^2 \right].$$

- The first term $\lim_{n \rightarrow \infty} n \text{Var}(\hat{\theta}_n)$ is called the **asymptotic variance**.
- The second term $\lim_{n \rightarrow \infty} n \text{Bias}(\hat{\theta}_n)^2$ is the **asymptotic bias squared**.

Thus AMSE measures the long-run (large- n) accuracy of an estimator, balancing both variance and bias contributions.

Interpretation

- If $\hat{\theta}_n$ is *asymptotically unbiased* (i.e. $\text{bias} = o(n^{-1/2})$), then the bias term vanishes and AMSE is determined entirely by the asymptotic variance.
- For maximum likelihood estimators (MLEs), under regularity conditions,

$$\sqrt{n}(\hat{\theta}_n - \theta) \Rightarrow N(0, I(\theta)^{-1}),$$

where $I(\theta)$ is the Fisher information. Hence,

$$\text{AMSE}(\hat{\theta}_n) = I(\theta)^{-1}.$$

- For biased estimators, AMSE captures the trade-off: even with larger samples, persistent bias contributes to error.

Example 3.1.1

Suppose $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} N(\mu, \sigma^2)$ and $\hat{\mu} = \bar{X}_n$.

$$\text{Var}(\bar{X}_n) = \frac{\sigma^2}{n}, \quad \text{Bias}(\bar{X}_n) = 0.$$

Hence

$$n \cdot \text{MSE}(\bar{X}_n) = n \cdot \frac{\sigma^2}{n} = \sigma^2.$$

Therefore,

$$\text{AMSE}(\bar{X}_n) = \sigma^2.$$

Example 3.1.2

Suppose we estimate μ by

$$\hat{\mu}_n = \bar{X}_n + \frac{1}{\sqrt{n}},$$

where $\bar{X}_n$ is the sample mean.

- $\text{Var}(\hat{\mu}_n) = \text{Var}(\bar{X}_n) = \frac{\sigma^2}{n}$.
- $\text{Bias}(\hat{\mu}_n) = \frac{1}{\sqrt{n}}$.

Thus

$$n \cdot \text{MSE}(\hat{\mu}_n) = n \left(\frac{\sigma^2}{n} + \frac{1}{n} \right) = \sigma^2 + 1.$$

So

$$\text{AMSE}(\hat{\mu}_n) = \sigma^2 + 1.$$

Here, although the bias vanishes as $n \rightarrow \infty$, it is of order $1/\sqrt{n}$, so its contribution persists in the AMSE.

Key Takeaways

- AMSE summarizes the large-sample accuracy of estimators, combining variance and bias.
- For asymptotically unbiased estimators, AMSE coincides with asymptotic variance.
- MLEs typically achieve the lowest possible AMSE, given by the Cramér–Rao information bound.
- Comparing AMSE values is a standard way to compare competing estimators asymptotically.

3.2 Consistency

Definition 3.3 (Consistency). A sequence of estimators $\hat{\theta}_n$ is said to be consistent for the parameter θ if, as the sample size grows, the estimator converges in probability to the true parameter value:

$$\hat{\theta}_n \xrightarrow{p} \theta \quad \text{as } n \rightarrow \infty.$$

That is, for any $\varepsilon > 0$,

$$\lim_{n \rightarrow \infty} \mathbb{P}(|\hat{\theta}_n - \theta| > \varepsilon) = 0.$$

Interpretation. Consistency ensures that with enough data, the estimator gets arbitrarily close to the true parameter. In practice, a consistent estimator improves as the sample size grows and does not “wander off” from the truth.

Markov's inequality proof

Theorem 3.1 (Markov's inequality (continuous case)). Let X be any positive continuous random variable. Then for any $a > 0$,

$$\mathbb{P}(X \geq a) \leq \frac{\mathbb{E}[X]}{a}.$$

Proof. We compute the expectation:

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} x f_X(x) dx = \int_0^{\infty} x f_X(x) dx \quad (\text{since } X \text{ is positive-valued}).$$

Now restrict to the region $x \geq a$:

$$\mathbb{E}[X] \geq \int_a^{\infty} x f_X(x) dx.$$

On this region we have $x \geq a$, hence

$$\int_a^{\infty} x f_X(x) dx \geq \int_a^{\infty} a f_X(x) dx = a \int_a^{\infty} f_X(x) dx = a \mathbb{P}(X \geq a).$$

Therefore,

$$\mathbb{E}[X] \geq a \mathbb{P}(X \geq a).$$

Rearranging gives the inequality:

$$\mathbb{P}(X \geq a) \leq \frac{\mathbb{E}[X]}{a}, \quad a > 0.$$

The same argument extends to discrete or mixed random variables by replacing the integral with the corresponding sum or generalized expectation. This result is known as *Markov's inequality*. $\square$

Proof of Chebyshev inequality

Theorem 3.2 (Chebyshev's inequality). *Let X be a random variable with finite mean $\mu = \mathbb{E}[X]$ and variance $\sigma^2 = \text{Var}(X) < \infty$. Then for every $\varepsilon > 0$,*

$$\mathbb{P}(|X - \mu| \geq \varepsilon) \leq \frac{\sigma^2}{\varepsilon^2}.$$

Proof via Markov's inequality. Recall Markov's inequality: if $Y \geq 0$ and $a > 0$, then $\mathbb{P}(Y \geq a) \leq \frac{\mathbb{E}[Y]}{a}$. Apply it with the nonnegative random variable $Y = (X - \mu)^2$ and $a = \varepsilon^2$:

$$\mathbb{P}(|X - \mu| \geq \varepsilon) = \mathbb{P}((X - \mu)^2 \geq \varepsilon^2) \leq \frac{\mathbb{E}[(X - \mu)^2]}{\varepsilon^2} = \frac{\sigma^2}{\varepsilon^2}.$$

□

3.2.1 Sufficient condition (Chebyshev).

A simple condition for consistency is based on the bias and variance of an estimator:

- If the **bias** vanishes as $n \rightarrow \infty$:

$$\text{Bias}(\hat{\theta}_n) = \mathbb{E}[\hat{\theta}_n] - \theta \rightarrow 0,$$

- and the **variance** goes to zero:

$$\text{Var}(\hat{\theta}_n) \rightarrow 0,$$

then $\hat{\theta}_n$ is consistent for θ .

Indeed, recall that

$$\text{MSE}(\hat{\theta}_n) = \text{Var}(\hat{\theta}_n) + (\text{Bias}(\hat{\theta}_n))^2.$$

If both terms vanish, then $\text{MSE}(\hat{\theta}_n) \rightarrow 0$. By Chebyshev's inequality,

$$\mathbb{P}(|\hat{\theta}_n - \theta| > \varepsilon) \leq \frac{\text{MSE}(\hat{\theta}_n)}{\varepsilon^2} \rightarrow 0,$$

which is exactly the definition of consistency.

Example 3.2.1

[Sample Mean] Suppose $X_1, X_2, \dots, X_n$ are i.i.d. with mean μ and variance $\sigma^2 < \infty$. The sample mean is

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i.$$

- **Expectation:**

$$\mathbb{E}[\bar{X}_n] = \mu \implies \text{Bias}(\bar{X}_n) = 0.$$

- **Variance:**

$$\text{Var}(\bar{X}_n) = \frac{\sigma^2}{n} \rightarrow 0 \quad \text{as } n \rightarrow \infty.$$

Thus both conditions are satisfied. Therefore,

$$\bar{X}_n \xrightarrow{p} \mu,$$

which is a direct application of the Weak Law of Large Numbers (WLLN).

Intuition.

- Consistency means “more data = better estimates.”
- A biased estimator can still be consistent if the bias disappears as n grows.
- An unbiased estimator may fail to be consistent if its variance does not shrink.
- Consistency is about *long-run accuracy*, not finite-sample performance.

3.3 Efficiency

Definition 3.4 (Finite-sample efficiency). *Suppose $\mathcal{U}$ is the class of all unbiased estimators of a parameter θ . An estimator $\hat{\theta} \in \mathcal{U}$ is said to be (uniformly) efficient if, for every other unbiased estimator $\tilde{\theta} \in \mathcal{U}$ and for every value of θ ,*

$$\text{Var}_{\theta}(\hat{\theta}) \leq \text{Var}_{\theta}(\tilde{\theta}).$$

Interpretation. Efficiency in the finite-sample sense means that $\hat{\theta}$ makes the most precise use of the available data among all unbiased competitors. If such an estimator exists, it achieves the lowest variance possible for unbiased estimation.

3.3.1 Asymptotic Relative Efficiency (ARE)

The **Asymptotic Relative Efficiency (ARE)** is a measure used to compare the efficiency of two estimators when the sample size becomes very large.

Suppose $\hat{\theta}$ and $\tilde{\theta}$ are two estimators of the same parameter θ . The asymptotic relative efficiency of $\hat{\theta}$ with respect to $\tilde{\theta}$ is defined as

$$\text{ARE}(\hat{\theta}, \tilde{\theta}) = \frac{\text{Asymptotic variance of } \tilde{\theta}}{\text{Asymptotic variance of } \hat{\theta}}.$$

Equivalently, if

$$\text{Var}_{\infty}(\hat{\theta}) = \sigma_{\hat{\theta}}^2 \quad \text{and} \quad \text{Var}_{\infty}(\tilde{\theta}) = \sigma_{\tilde{\theta}}^2,$$

then

$$\text{ARE}(\hat{\theta}, \tilde{\theta}) = \frac{\sigma_{\tilde{\theta}}^2}{\sigma_{\hat{\theta}}^2}.$$

Interpretation The value of ARE indicates which estimator is more efficient as the sample size becomes large.

- If $\text{ARE}(\hat{\theta}, \tilde{\theta}) > 1$, then $\hat{\theta}$ is asymptotically more efficient than $\tilde{\theta}$, since it has a smaller asymptotic variance.
- If $\text{ARE}(\hat{\theta}, \tilde{\theta}) < 1$, then $\tilde{\theta}$ is more efficient.
- If $\text{ARE}(\hat{\theta}, \tilde{\theta}) = 1$, the two estimators are asymptotically equivalent in efficiency.

Example 3.3.1

Consider estimating the mean μ of a normal distribution using the sample mean $\bar{X}$ and the sample median $\tilde{X}$.
For normally distributed data,

$$\text{ARE}(\tilde{X}, \bar{X}) = \frac{2}{\pi} \approx 0.637.$$

Thus, the sample median has approximately 63.7% of the efficiency of the sample mean when the data follow a normal distribution.

- If $\text{ARE}(\hat{\theta}, \tilde{\theta}) > 1$, then $\hat{\theta}$ is asymptotically more efficient than $\tilde{\theta}$, since it has a smaller asymptotic variance.
- If $\text{ARE}(\hat{\theta}, \tilde{\theta}) < 1$, then $\tilde{\theta}$ is more efficient.
- If $\text{ARE} = 1$, the two estimators are asymptotically equivalent in efficiency.

Interpretation: The sample mean is about 57% more efficient than the median under the normal distribution. Equivalently, the median would require roughly 1.57 times as many observations to achieve the same precision as the mean.

Cauchy–Schwarz Inequality

Theorem 3.3 (Cauchy–Schwarz inequality). *Let X, Y be square-integrable random variables. Then*

$$\text{Cov}(X, Y)^2 \leq \text{Var}(X) \text{Var}(Y).$$

Equivalently, the correlation satisfies $|\rho_{X,Y}| \leq 1$, where $\rho_{X,Y} = \text{Cov}(X, Y) / \sqrt{\text{Var}(X) \text{Var}(Y)}$.

Proof by nonnegativity of variance. For any real t , the variance of $X - tY$ is non-negative:

$$0 \leq \text{Var}(X - tY).$$

Expand using bilinearity of covariance and $\text{Var}(Z) = \text{Cov}(Z, Z)$:

$$\text{Var}(X - tY) = \text{Var}(X) - 2t \text{Cov}(X, Y) + t^2 \text{Var}(Y).$$

Viewed as a quadratic polynomial in t ,

$$q(t) = \text{Var}(Y) t^2 - 2 \text{Cov}(X, Y) t + \text{Var}(X),$$

the condition $\text{Var}(X - tY) \geq 0$ for all $t \in \mathbb{R}$ implies that $q(t) \geq 0$ for all t . Hence its discriminant is nonpositive:

$$\Delta = (-2 \text{Cov}(X, Y))^2 - 4 \text{Var}(Y) \text{Var}(X) \leq 0.$$

Rearranging gives

$$\text{Cov}(X, Y)^2 \leq \text{Var}(X) \text{Var}(Y).$$

□

3.3.2 Fisher Information

When we estimate a parameter θ from data, not all samples are equally “informative.” Fisher information quantifies how much information an observable random variable carries about an unknown parameter. It is central to the Cramér–Rao lower bound (CRLB), which limits the precision of unbiased estimators.

Definition 3.5 (One variable). *Let X be a random variable with density (or pmf) $f(x; \theta)$ depending on parameter θ . The **score function** (derivative of the log-likelihood with respect to the parameter) is*

$$U(X; \theta) = \frac{\partial}{\partial \theta} \log f(X; \theta).$$

The **Fisher information** is defined as

$$I(\theta) = \mathbb{E}_\theta [U(X; \theta)^2] = \mathbb{E}_\theta \left[\left(\frac{\partial}{\partial \theta} \log f(X; \theta) \right)^2 \right].$$

Equivalently, under regularity conditions,

$$I(\theta) = -\mathbb{E}_\theta \left[\frac{\partial^2}{\partial \theta^2} \log f(X; \theta) \right].$$

Definition 3.6 (Sample of Size n). *For $X_1, \dots, X_n$ i.i.d. $\sim f(x; \theta)$, the Fisher information in the full sample is additive:*

$$I_n(\theta) = n \cdot I(\theta).$$

Interpretation

- $I(\theta)$ measures the *curvature* of the log-likelihood at the true parameter value.

- A flatter log-likelihood curve means less information (estimation is harder).
- A sharper curve means more information (estimation is easier).
- High Fisher information $\Rightarrow$ data are very informative about θ , so estimators can be more precise.
- Low Fisher information $\Rightarrow$ data carry little information, so estimation must be imprecise.

Example 3.3.2

[Normal Mean (Variance Known).] Suppose $X \sim N(\mu, \sigma^2)$ with σ^2 known.

$$\log f(x; \mu) = -\frac{1}{2} \log(2\pi\sigma^2) - \frac{(x-\mu)^2}{2\sigma^2}.$$

$$\frac{\partial}{\partial \mu} \log f(x; \mu) = \frac{x - \mu}{\sigma^2}.$$

$$I(\mu) = \mathbb{E} \left[\left(\frac{X - \mu}{\sigma^2} \right)^2 \right].$$

We want to evaluate

$$\mathbb{E} \left[\left(\frac{X - \mu}{\sigma^2} \right)^2 \right],$$

where $X \sim N(\mu, \sigma^2)$ with μ unknown and σ^2 known.
Since σ^2 is not random,

$$\mathbb{E} \left[\left(\frac{X - \mu}{\sigma^2} \right)^2 \right] = \frac{1}{\sigma^4} \mathbb{E}[(X - \mu)^2].$$

By definition of variance,

$$\text{Var}(X) = \mathbb{E}[(X - \mu)^2].$$

For $X \sim N(\mu, \sigma^2)$,

$$\mathbb{E}[(X - \mu)^2] = \sigma^2.$$

Hence

$$\frac{1}{\sigma^4} \mathbb{E}[(X - \mu)^2] = \frac{1}{\sigma^4} \cdot \sigma^2 = \frac{1}{\sigma^2}.$$

Therefore,

$$\mathbb{E} \left[\left(\frac{X - \mu}{\sigma^2} \right)^2 \right] = \frac{1}{\sigma^2}.$$

This shows that the Fisher information for a single Normal observation with mean μ (and known variance σ^2) is

$$I(\mu) = \frac{1}{\sigma^2}.$$

So n observations give $I_n(\mu) = \frac{n}{\sigma^2}$.

Example 3.3.3

Let $X \sim \text{Bernoulli}(p)$ with $0 < p < 1$. Its pmf is

$$f(x; p) = p^x (1 - p)^{1-x}, \quad x \in \{0, 1\}.$$

$$\ell(p; x) = \log f(x; p) = x \log p + (1 - x) \log(1 - p).$$

$$U(x; p) \equiv \frac{\partial}{\partial p} \ell(p; x) = \frac{x}{p} - \frac{1-x}{1-p}.$$

Check that the score has mean zero

$$\mathbb{E}_p[U(X; p)] = \mathbb{E}\left[\frac{X}{p} - \frac{1-X}{1-p}\right] = \frac{\mathbb{E}[X]}{p} - \frac{1 - \mathbb{E}[X]}{1-p} = \frac{p}{p} - \frac{1-p}{1-p} = 1 - 1 = 0.$$

By definition (one observation),

$$I(p) = \mathbb{E}_p[U(X; p)^2] = \sum_{x \in \{0,1\}} U(x; p)^2 f(x; p).$$

Evaluate $U(x; p)$ at $x = 1$ and $x = 0$:

$$U(1; p) = \frac{1}{p} - 0 = \frac{1}{p}, \quad U(0; p) = 0 - \frac{1}{1-p} = -\frac{1}{1-p}.$$

Hence

$$I(p) = \left(\frac{1}{p}\right)^2 f(1; p) + \left(-\frac{1}{1-p}\right)^2 f(0; p) = \frac{1}{p^2} p + \frac{1}{(1-p)^2} (1-p) = \frac{1}{p} + \frac{1}{1-p}.$$

Combine the terms:

$$I(p) = \frac{1-p+p}{p(1-p)} = \frac{1}{p(1-p)}.$$

Additivity for n i.i.d. trials. For $X_1, \dots, X_n$ i.i.d. $\text{Bernoulli}(p)$, the sample information is

$$I_n(p) = n I(p) = \frac{n}{p(1-p)}.$$

3.3.3 The Cramér–Rao Lower Bound (CRLB)

In estimation theory, one of the key questions is: *How precise can an unbiased estimator possibly be?*

The Cramér–Rao Lower Bound (CRLB) provides a universal lower bound on the variance of any unbiased estimator of a parameter θ . This bound is determined entirely by the probability model through the Fisher information.

Theorem 3.4 (Cramér–Rao Lower Bound (CRLB)). *Let $X_1, \dots, X_n$ be i.i.d. with density*

(or pmf) $f(x; \theta)$, satisfying standard regularity conditions (differentiability, support not depending on θ , interchange of differentiation and expectation).

Then for any unbiased estimator $\hat{\theta}$ of θ ,

$$\text{Var}_{\theta}(\hat{\theta}) \geq \frac{1}{I_n(\theta)} = \frac{1}{nI(\theta)}.$$

Moreover, equality holds if and only if (a.s.)

$$\hat{\theta} - \theta = c(\theta) \cdot \sum_{i=1}^n \frac{\partial}{\partial \theta} \log f(X_i; \theta),$$

for some function $c(\theta)$, i.e. when the estimator is linear in the score function.

—

Efficiency. An unbiased estimator $\hat{\theta}$ is called *efficient* if it achieves the CRLB, i.e.

$$\text{Var}(\hat{\theta}) = \frac{1}{nI(\theta)}.$$

Notation. Write the sample log-likelihood and sample score as

$$\ell_n(\theta) = \sum_{i=1}^n \log f(X_i; \theta), \quad U_n(\theta) = \ell'_n(\theta) = \sum_{i=1}^n \frac{\partial}{\partial \theta} \log f(X_i; \theta).$$

Proof. Step 1: Differentiate unbiasedness and express as a covariance. Unbiasedness gives $\mathbb{E}_{\theta}[\hat{\theta}] = \theta$. Differentiate w.r.t. θ :

$$\frac{\partial}{\partial \theta} \mathbb{E}_{\theta}[\hat{\theta}] = 1.$$

with $f(x; \theta)$ is differentiable in θ , and differentiation under the integral sign is justified:

$$\frac{\partial}{\partial \theta} \mathbb{E}_{\theta}[h(X_1, \dots, X_n)] = \mathbb{E}_{\theta} \left[h(X_1, \dots, X_n) \frac{\partial}{\partial \theta} \log f(X_1, \dots, X_n; \theta) \right],$$

for appropriate integrable h . bring the derivative inside the expectation and use the identity $\frac{\partial}{\partial \theta} \log f(X_1, \dots, X_n; \theta) = U_n(\theta)$:

$$\frac{\partial}{\partial \theta} \mathbb{E}_{\theta}[\hat{\theta}] = \mathbb{E}_{\theta}[\hat{\theta} U_n(\theta)] = 1.$$

we identify by definition,

$$\text{Cov}_{\theta}(\hat{\theta}, U_n(\theta)) = \mathbb{E}_{\theta} \left[(\hat{\theta} - \mathbb{E}_{\theta}[\hat{\theta}]) (U_n(\theta) - \mathbb{E}_{\theta}[U_n(\theta)]) \right].$$

$$\text{Cov}_{\theta}(\hat{\theta}, U_n(\theta)) = \mathbb{E}_{\theta}[\hat{\theta} U_n(\theta)] - \mathbb{E}_{\theta}[\hat{\theta}] \mathbb{E}_{\theta}[U_n(\theta)].$$

But we know:

$$\mathbb{E}_\theta[\hat{\theta}] = \theta \quad (\text{unbiasedness}), \quad \mathbb{E}_\theta[U_n(\theta)] = 0 \quad (\text{property of the score}).$$

$$\mathbb{E}_\theta[U_n(\theta)] = \int \frac{\partial}{\partial \theta} \log f(x; \theta) f(x; \theta) dx = \int \frac{\partial}{\partial \theta} f(x; \theta) dx = \frac{\partial}{\partial \theta} \int f(x; \theta) dx = 0.$$

$$\mathbb{E}_\theta[U_n(\theta)] = \int \frac{\partial}{\partial \theta} \log f(x; \theta) f(x; \theta) dx.$$

Note that

$$\frac{\partial}{\partial \theta} \log f(x; \theta) = \frac{1}{f(x; \theta)} \cdot \frac{\partial}{\partial \theta} f(x; \theta).$$

Therefore,

$$\left(\frac{\partial}{\partial \theta} \log f(x; \theta) \right) f(x; \theta) = \frac{1}{f(x; \theta)} \frac{\partial}{\partial \theta} f(x; \theta) f(x; \theta) = \frac{\partial}{\partial \theta} f(x; \theta).$$

Hence,

$$\mathbb{E}_\theta[U_n(\theta)] = \int \frac{\partial}{\partial \theta} f(x; \theta) dx.$$

Since $f(x; \theta)$ is a probability density (or pmf),

$$\int f(x; \theta) dx = 1,$$

and thus

$$\mathbb{E}_\theta[U_n(\theta)] = \frac{\partial}{\partial \theta} \int f(x; \theta) dx = \frac{\partial}{\partial \theta} 1 = 0.$$

Therefore,

$$\text{Cov}_\theta(\hat{\theta}, U_n(\theta)) = \mathbb{E}_\theta[\hat{\theta} U_n(\theta)] = 1$$

Apply Cauchy-Schwarz. By Cauchy–Schwarz,

$$(\text{Cov}_\theta(\hat{\theta}, U_n)) ^2 \leq \text{Var}_\theta(\hat{\theta}) \text{Var}_\theta(U_n(\theta)).$$

But $\text{Var}_\theta(U_n(\theta)) = \mathbb{E}_\theta(U_n(\theta)^2) - (\mathbb{E}_\theta(U_n(\theta)))^2 = \mathbb{E}_\theta(U_n(\theta)^2) = I_n(\theta)$:

$$1^2 \leq \text{Var}_\theta(\hat{\theta}) I_n(\theta).$$

Hence

$$\text{Var}_\theta(\hat{\theta}) \geq \frac{1}{I_n(\theta)}.$$

Step 3: Equality condition. Equality in Cauchy–Schwarz holds iff $\hat{\theta} - \mathbb{E}_\theta[\hat{\theta}]$ is a.s. proportional to $U_n(\theta)$. Since $\mathbb{E}_\theta[\hat{\theta}] = \theta$,

$$\hat{\theta} - \theta = c(\theta) U_n(\theta) \quad \text{a.s.}$$

for some (measurable) $c(\theta)$. This is the stated condition. □

Generalization to functions of θ . Let $g : \Theta \rightarrow \mathbb{R}$ be differentiable and suppose T is an unbiased estimator of $g(\theta)$: $\mathbb{E}_\theta[T] = g(\theta)$. Repeating the steps with $g'(\theta)$ replacing 1, one obtains the *information inequality*

$$\text{Var}_\theta(T) \geq \frac{(g'(\theta))^2}{I_n(\theta)}.$$

Alternative form via the expected Hessian. Under (R2)–(R3), $I(\theta) = -\mathbb{E}_\theta[\ell''(\theta; X)]$ and hence $I_n(\theta) = -\mathbb{E}_\theta[\ell''_n(\theta)]$. Thus the bound reads $\text{Var}_\theta(\hat{\theta}) \geq (-\mathbb{E}_\theta[\ell''_n(\theta)])^{-1}$.

Remarks.

- The CRLB is a bound for *unbiased* estimators. For biased estimators, a generalized bound (Bhattacharyya/biased CRLB) includes derivatives of the bias.
- Maximum likelihood estimators (under regularity) satisfy $\sqrt{n}(\hat{\theta}_{\text{MLE}} - \theta) \Rightarrow N(0, I(\theta)^{-1})$, hence are *asymptotically efficient*, achieving the bound in the limit.
- If the support depends on θ or regularity fails, the bound may not hold or may need modification.

3.4 Sufficiency

Definition 3.7 (Sufficient Statistic). *A statistic $T(X)$ is said to be sufficient for a parameter θ if the conditional distribution of the sample X given $T(X)$ does not depend on θ .*

Equivalently, once $T(X)$ is known, the data X provide no additional information about θ .

Intuition. Sufficiency captures the idea of *data reduction without loss of information about the parameter*. That is, $T(X)$ summarizes all the information in the sample relevant to estimating θ . For example, in the normal model $N(\mu, \sigma^2)$ with known σ^2 , the sample mean $\bar{X}$ is sufficient for μ — no other function of the data can improve our knowledge about μ .

Theorem 3.5 (Neyman–Fisher Factorization Criterion). *Let $X_1, \dots, X_n$ have joint density or mass function $f(x_1, \dots, x_n; \theta)$. A statistic $T(X)$ is sufficient for θ if and only if the likelihood can be factorized as*

$$f(x; \theta) = g_\theta(T(x)) h(x),$$

where g_θ depends on x only through $T(x)$, and $h(x)$ is free of θ .

3.4.1 Proof of the Neyman–Fisher Factorization Theorem.

We show that if the joint pdf or pmf $f(x; \theta)$ can be written as

$$f(x; \theta) = g_\theta(T(x)) h(x),$$

for some nonnegative functions g_θ and h (with h independent of θ), then $T(X)$ is a **sufficient statistic** for θ .

Step 1. Recall the definition of sufficiency:

$$T(X) \text{ is sufficient for } \theta \iff f(x \mid T(X) = t; \theta) \text{ does not depend on } \theta.$$

That is, once $T(X)$ is known, the conditional distribution of the full sample X does not contain further information about θ .

Step 2. Write down the joint density $f(x; \theta)$ and the marginal density of $T(X)$, denoted by $f_T(t; \theta)$.

For discrete data:

$$f_T(t; \theta) = \sum_{x': T(x')=t} f(x'; \theta),$$

and for continuous data:

$$f_T(t; \theta) = \int_{\{x': T(x')=t\}} f(x'; \theta) dx'.$$

Step 3. Express the conditional distribution of X given $T(X) = t$. For discrete data:

$$f(x \mid T(X) = t; \theta) = \frac{f(x; \theta)}{f_T(t; \theta)} = \frac{f(x; \theta)}{\sum_{x': T(x')=t} f(x'; \theta)}.$$

This ratio represents how the probability mass of X is distributed among all data points having the same sufficient statistic value t .

Step 4. Substitute the factorized form of the likelihood:

$$f(x; \theta) = g_\theta(T(x)) h(x).$$

For every x such that $T(x) = t$, we have

$$f(x; \theta) = g_\theta(t) h(x).$$

Then the conditional pmf becomes

$$f(x \mid T(X) = t; \theta) = \frac{g_\theta(t) h(x)}{\sum_{x': T(x')=t} g_\theta(t) h(x')} = \frac{h(x)}{\sum_{x': T(x')=t} h(x')}.$$

Step 5. Observe that in this expression, the term $g_\theta(t)$ cancels out completely — the conditional distribution no longer depends on θ .

$$f(x \mid T(X) = t; \theta) = \frac{h(x)}{\sum_{x': T(x')=t} h(x')}.$$

Hence, once $T(X)$ is known, the conditional distribution of X given $T(X) = t$ is independent of θ .

Step 6. Conclusion. By the definition of sufficiency, this proves that $T(X)$ is sufficient for θ .

$$\boxed{T(X) \text{ is sufficient for } \theta.}$$

Continuous Case (analogous argument). If X has pdf $f(x; \theta) = g_\theta(T(x)) h(x)$, then

$$f(x \mid T(X) = t; \theta) = \frac{f(x; \theta)}{\int_{T(x')=t} f(x'; \theta) dx'} = \frac{h(x)}{\int_{T(x')=t} h(x') dx'},$$

which again is free of θ . Therefore, $T(X)$ is sufficient.

Interpretation. The factorization theorem formalizes the intuition that the sample X contains information about θ only through $T(X)$. Once $T(X)$ is known, the rest of the data pattern (i.e., the configuration of X values consistent with $T(X) = t$) is purely random and provides no further knowledge about θ .

The converse follows by the same argument.

—

Example 3.4.1

[Bernoulli Family] Suppose $X_1, X_2, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} \text{Bernoulli}(p)$, where each X_i can take values 0 or 1, with

$$P(X_i = 1) = p, \quad P(X_i = 0) = 1 - p, \quad 0 < p < 1.$$

Step 1. Joint Probability Mass Function. Because the sample is independent and identically distributed, the joint pmf of the sample $x = (x_1, x_2, \dots, x_n)$ is

$$f(x; p) = \prod_{i=1}^n p^{x_i} (1 - p)^{1-x_i}.$$

Expanding this product gives

$$f(x; p) = p^{x_1+x_2+\dots+x_n} (1 - p)^{n-(x_1+x_2+\dots+x_n)}.$$

Define the statistic

$$T(X) = \sum_{i=1}^n X_i,$$

which counts the total number of successes in the sample.

Then the joint pmf can be compactly written as

$$f(x; p) = p^{T(x)} (1 - p)^{n-T(x)}.$$

Step 2. Identify the Factorization. We can express this pmf as

$$f(x; p) = \underbrace{p^{T(x)}(1-p)^{n-T(x)}}_{g_p(T(x))} \underbrace{1}_{h(x)}.$$

Here:

- $g_p(T(x)) = p^{T(x)}(1-p)^{n-T(x)}$ depends on the data only through the statistic $T(x)$ and on the parameter p ,
- $h(x) = 1$ is a function of the data only and does not involve p .

This is precisely in the Neyman–Fisher factorization form

$$f(x; p) = g_p(T(x)) h(x).$$

Hence, by the factorization theorem, $T(X) = \sum_{i=1}^n X_i$ is a **sufficient statistic** for p .

Step 3. Verify Using Conditional Distribution (optional check). We can verify sufficiency directly. The conditional pmf of $X = (X_1, \dots, X_n)$ given $T(X) = t$ is

$$f(x \mid T(X) = t; p) = \frac{f(x; p)}{\sum_{x': T(x')=t} f(x'; p)}.$$

For all x such that $T(x) = t$,

$$f(x; p) = p^t(1-p)^{n-t}.$$

Hence,

$$f(x \mid T(X) = t; p) = \frac{p^t(1-p)^{n-t}}{\sum_{x': T(x')=t} p^t(1-p)^{n-t}} = \frac{1}{\#\{x' : T(x') = t\}} = \frac{1}{\binom{n}{t}}.$$

The denominator counts the number of sample configurations with exactly t successes, i.e. $\binom{n}{t}$. Thus, this conditional pmf is *independent of p* , confirming that $T(X)$ is indeed sufficient for p .

Step 4. Intuition. The statistic $T(X)$ records only the total number of successes. Given $T(X) = t$, all specific arrangements of t ones and $n - t$ zeros in the data have equal probability. Therefore, the order or pattern of successes does not contain any extra information about p beyond the total count t . All inference about p (such as MLE or confidence intervals) depends solely on $T(X)$.

Step 5. Connection to Other Results. The sufficiency of $T(X)$ also explains why:

- The likelihood function depends on p only through $T(X)$:

$$L(p; x) = p^{T(x)}(1-p)^{n-T(x)};$$

- The MLE of p ,

$$\hat{p} = \frac{T(X)}{n} = \bar{X},$$

depends only on the sufficient statistic $T(X)$.

Thus, $T(X)$ achieves complete data reduction for inference about p .

Conclusion.

$$T(X) = \sum_{i=1}^n X_i \text{ is sufficient for } p \text{ in the Bernoulli}(p) \text{ model.}$$

Example 3.4.2

[Normal Family: Joint Sufficiency of $(\bar{X}, S^2)$] Let $X_1, X_2, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} N(\mu, \sigma^2)$, where both μ and σ^2 are unknown parameters.

Step 1. Write the joint probability density. The pdf of a single observation is

$$f(x_i; \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{(x_i - \mu)^2}{2\sigma^2}\right].$$

Since the X_i are independent,

$$f(x_1, \dots, x_n; \mu, \sigma^2) = \prod_{i=1}^n f(x_i; \mu, \sigma^2) = (2\pi\sigma^2)^{-n/2} \exp\left[-\frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2\right].$$

This is the joint likelihood function:

$$L(\mu, \sigma^2; x) = (2\pi\sigma^2)^{-n/2} \exp\left[-\frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2\right].$$

Step 2. Expand the quadratic term. We decompose the sum of squared deviations into two parts: variation due to the mean difference $(\bar{X} - \mu)$ and variation within the sample.

$$\sum_{i=1}^n (x_i - \mu)^2 = \sum_{i=1}^n [(x_i - \bar{X}) + (\bar{X} - \mu)]^2.$$

Expanding:

$$\sum (x_i - \mu)^2 = \sum (x_i - \bar{X})^2 + 2(\bar{X} - \mu) \sum (x_i - \bar{X}) + n(\bar{X} - \mu)^2.$$

The middle term vanishes because $\sum (x_i - \bar{X}) = 0$. Hence,

$$\sum (x_i - \mu)^2 = \sum (x_i - \bar{X})^2 + n(\bar{X} - \mu)^2.$$

Step 3. Substitute back into the likelihood.

$$L(\mu, \sigma^2; x) = (2\pi\sigma^2)^{-n/2} \exp\left[-\frac{1}{2\sigma^2} \left\{ n(\bar{X} - \mu)^2 + \sum (x_i - \bar{X})^2 \right\}\right].$$

Rearranging terms:

$$L(\mu, \sigma^2; x) = \underbrace{(2\pi\sigma^2)^{-n/2} \exp\left[-\frac{n(\bar{X} - \mu)^2}{2\sigma^2}\right]}_{\text{joint pdf written for factorization}} \exp\left[-\frac{\sum (x_i - \bar{X})^2}{2\sigma^2}\right].$$

Step 4. Apply the Neyman–Fisher Factorization Theorem. The likelihood can be rewritten as

$$L(\mu, \sigma^2; x) = \underbrace{(2\pi\sigma^2)^{-n/2} \exp\left[-\frac{n(\bar{X} - \mu)^2}{2\sigma^2}\right] \exp\left[-\frac{\sum (x_i - \bar{X})^2}{2\sigma^2}\right]}_{=g_{(\mu, \sigma^2)}(\bar{X}, S^2) h(x)}.$$

To make the dependence clearer, define

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{X})^2.$$

Then we can express the likelihood as:

$$L(\mu, \sigma^2; x) = \underbrace{(2\pi\sigma^2)^{-n/2} \exp\left[-\frac{n(\bar{X} - \mu)^2}{2\sigma^2}\right] \exp\left[-\frac{(n-1)S^2}{2\sigma^2}\right]}_{g_{(\mu, \sigma^2)}(\bar{X}, S^2)} \times \underbrace{1}_{h(x)}.$$

Here: - $g_{(\mu, \sigma^2)}(\bar{X}, S^2)$ depends on the sample only through the pair $(\bar{X}, S^2)$, and on the parameters (μ, σ^2) , - $h(x)$ does not depend on the parameters.

Therefore, by the Neyman–Fisher factorization theorem, $(\bar{X}, S^2)$ is a ****sufficient statistic**** for (μ, σ^2) .

Step 5. Interpretation.

- The data $x_1, \dots, x_n$ influence the likelihood only through $\bar{X}$ and S^2 .
- Once $(\bar{X}, S^2)$ are known, the remaining information in the sample (for example, the order of the x_i) provides no further knowledge about (μ, σ^2) .
- Hence, $(\bar{X}, S^2)$ provides complete information about both parameters — it compresses n data points into two statistics without losing inferential power.

Step 6. Further Insights.

1. **Marginal distributions:**

$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right), \quad \frac{(n-1)S^2}{\sigma^2} \sim \chi_{n-1}^2,$$

and $\bar{X}$ and S^2 are independent. This independence reinforces that together they capture all the information about (μ, σ^2) .

2. **Minimal sufficiency:** $(\bar{X}, S^2)$ is not only sufficient but also *minimal sufficient*, since any other sufficient statistic for (μ, σ^2) must be a function of $(\bar{X}, S^2)$.

3. **Connection to estimation:** The MLEs of μ and σ^2 ,

$$\hat{\mu} = \bar{X}, \quad \hat{\sigma}^2 = \frac{1}{n} \sum (X_i - \bar{X})^2,$$

are functions of the sufficient statistics.

Step 7. Visual/Intuitive Explanation. The sample mean $\bar{X}$ measures the location (central tendency), while S^2 measures the spread (variability). In the normal distribution, these two quantities completely determine the shape of the likelihood function over (μ, σ^2) . All other features of the sample are merely random fluctuations around these two core summaries.

Conclusion.

$(\bar{X}, S^2)$ is a jointly sufficient statistic for (μ, σ^2) in the $\text{Normal}(\mu, \sigma^2)$ family.

3.4.2 Minimal Sufficiency

A sufficient statistic $T(X)$ is said to be *minimal sufficient* if it is a function of every other sufficient statistic. That is, it achieves the greatest possible data reduction while retaining sufficiency.

Practical Test (Lehmann's Criterion). Two sample points x and y belong to the same equivalence class of the minimal sufficient statistic if

$$\frac{f(x; \theta)}{f(y; \theta)} \text{ is constant in } \theta.$$

Hence, the minimal sufficient statistic partitions the sample space into the largest possible sets for which this ratio is constant.

Example 3.4.3

[Normal Mean, Known Variance] Consider a random sample $X_1, X_2, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} N(\mu, 1)$, where μ is unknown and the variance is known ($\sigma^2 = 1$).

Step 1. Write down the joint pdf. The pdf of a single observation is

$$f(x_i; \mu) = \frac{1}{\sqrt{2\pi}} \exp \left[-\frac{(x_i - \mu)^2}{2} \right].$$

Hence, for a sample $x = (x_1, \dots, x_n)$, the joint pdf is

$$f(x; \mu) = (2\pi)^{-n/2} \exp \left[-\frac{1}{2} \sum_{i=1}^n (x_i - \mu)^2 \right].$$

Step 2. Recall Lehmann's Criterion for Minimal Sufficiency. A statistic $T(X)$ is **minimal sufficient** for μ if and only if for any two sample points x and y ,

$$\frac{f(x; \mu)}{f(y; \mu)} \text{ is constant in } \mu \iff T(x) = T(y).$$

This means that the ratio of their likelihoods must be free of μ if and only if the two samples yield the same statistic value.

Step 3. Compute the likelihood ratio. From the joint pdf:

$$f(x; \mu) = (2\pi)^{-n/2} \exp\left[-\frac{1}{2} \sum (x_i - \mu)^2\right], \quad f(y; \mu) = (2\pi)^{-n/2} \exp\left[-\frac{1}{2} \sum (y_i - \mu)^2\right].$$

Thus,

$$\frac{f(x; \mu)}{f(y; \mu)} = \exp\left\{-\frac{1}{2} \left[\sum (x_i - \mu)^2 - \sum (y_i - \mu)^2\right]\right\}.$$

Step 4. Expand both quadratic sums. We can expand each term:

$$\sum (x_i - \mu)^2 = \sum x_i^2 - 2\mu \sum x_i + n\mu^2, \quad \sum (y_i - \mu)^2 = \sum y_i^2 - 2\mu \sum y_i + n\mu^2.$$

Subtracting gives:

$$\sum (x_i - \mu)^2 - \sum (y_i - \mu)^2 = (\sum x_i^2 - \sum y_i^2) - 2\mu(\sum x_i - \sum y_i).$$

Step 5. Substitute back into the likelihood ratio.

$$\frac{f(x; \mu)}{f(y; \mu)} = \exp\left\{-\frac{1}{2} \left[(\sum x_i^2 - \sum y_i^2) - 2\mu(\sum x_i - \sum y_i)\right]\right\}.$$

Simplify:

$$\frac{f(x; \mu)}{f(y; \mu)} = \exp\left[-\frac{1}{2}(\sum x_i^2 - \sum y_i^2)\right] \exp\left[\mu(\sum x_i - \sum y_i)\right].$$

Step 6. Identify when this ratio is constant in μ . The first exponential term

$$\exp\left[-\frac{1}{2}(\sum x_i^2 - \sum y_i^2)\right]$$

is free of μ , but the second term

$$\exp\left[\mu(\sum x_i - \sum y_i)\right]$$

depends on μ unless $\sum x_i = \sum y_i$.

Hence, the ratio $\frac{f(x; \mu)}{f(y; \mu)}$ is constant in μ if and only if

$$\sum_{i=1}^n x_i = \sum_{i=1}^n y_i.$$

Step 7. Apply Lehmann's Criterion. By Lehmann's criterion, the equivalence classes of sample points x that yield the same value of $\sum x_i$ form the level sets of the minimal sufficient statistic.

Therefore, the minimal sufficient statistic is

$$T(X) = \sum_{i=1}^n X_i,$$

or equivalently, since $\bar{X} = \frac{1}{n} \sum X_i$,

$$T(X) = \bar{X}.$$

Step 8. Interpretation. Intuitively:

- The likelihood for μ depends on the data only through $\sum X_i$ (or $\bar{X}$).
- Two samples with the same total (or same mean) produce identical likelihood functions up to a constant scaling factor.
- Thus, $\bar{X}$ captures all the information about μ present in the sample — no further function of X can reduce the data without losing information about μ .

Step 9. Connection to Sufficiency. We already know from the factorization theorem that $\bar{X}$ is *sufficient* for μ :

$$L(\mu; x) = (2\pi)^{-n/2} \exp\left[-\frac{1}{2} \sum (x_i - \mu)^2\right] = \underbrace{\exp\left[-\frac{1}{2} \sum (x_i - \bar{X})^2\right]}_{h(x)} \times \underbrace{\exp\left[-\frac{n(\bar{X} - \mu)^2}{2}\right]}_{g_\mu(\bar{X})}.$$

Lehmann's criterion goes further — it confirms that this statistic is not only sufficient but also the *minimal sufficient* one.

Conclusion.

$$T(X) = \sum_{i=1}^n X_i \text{ (or equivalently } \bar{X}) \text{ is minimal sufficient for } \mu \text{ when } X_i \sim N(\mu, 1).$$